

Chapter 2
Data transmission



IN THIS CHAPTER YOU WILL:

- learn how data is broken down into packets to be transmitted, including the contents of each packet
- understand how data is transmitted using a range of different transmission methods
- be able to choose a data transmission method for a given scenario
- learn about the universal serial bus (USB) interface and how it is used to transmit data
- learn how errors can occur when transmitting data
- be able to describe how a range of different error detection methods work
- understand why it is useful to encrypt data when it is transmitted
- be able to describe how data is encrypted using both symmetric and asymmetric encryption.

GETTING STARTED

Your data is transmitted to other devices, sometimes all over the world, on a regular basis. Have you thought about how it gets there? Draw a diagram to represent how you think data travels from your device to your friend's device when you send them a message. It doesn't matter if you don't think you know how this is done, just draw what you think happens and you can see at the end of the chapter how close you were with your diagram.

DATA TRANSMISSION IN THE FINANCIAL WORLD

A bank has to transmit data all around the world for thousands, possibly millions, of people on a daily basis. The accuracy of this data is imperative. Imagine what would happen if one person's hard earned money went into another person's bank account! When data is transmitted, there are certain things that can cause it to suffer errors. Interference is a big issue in data transmission that can cause errors in the data, therefore banks have to make sure that they have many error detection methods in place to make sure that data is accurate, even after transmission. As the accuracy of the data is so crucial for banking transactions, banks have to combine several types of error detection, to make sure that the data has been thoroughly and completely analysed for errors.

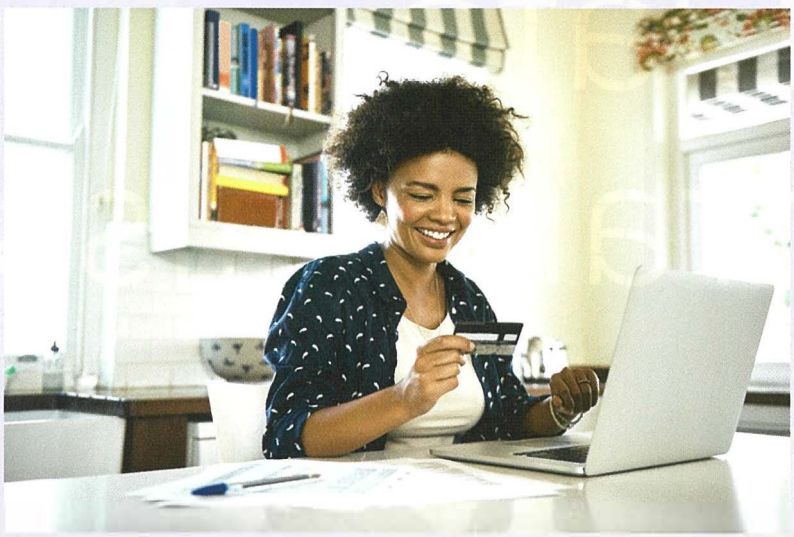


Figure 2.1: Financial transactions can be done online

The importance of checking for errors in data transmission can costs banks a lot of money. It takes very large and powerful systems to check the millions of transactions that are made every day. However, it would cost them even more in compensation money, if they messed up the data of their customers, and possibly their money, so they put a lot of time, money and effort into the process.

Discussion questions

1 How would you feel about having a bank account with a bank that did not have any error detection systems in place? What problems do you think this could cause?

2 Do you think it is right that the bank is responsible for checking errors in the transmission of data? Much of the hardware, that will be used to transmit the data from one device to another, will not be owned by them. So, why do you think the bank should be made responsible for checking for errors if the errors may occur using hardware that isn't theirs?

2.1 The structure of a data packet

The amount of data that is stored in a file can be very large. If you tried to transmit this data from one device to another, all at once, this just wouldn't be possible or practical. The wires or radio waves used would simply not be able to accommodate sending such large amounts of data at a single time. Therefore, for the data in a file to be transmitted, it is broken down into very small units called **packets**.

Each packet of data contains three different sections:

Packet header
Payload
Trailer

The **packet header** contains a lot of important information about the data enclosed in the packet and its destination. The information it includes is the:

- destination address
- packet number
- originators address.

The destination address is normally an **IP address**. It is the IP address of the device where the data is being sent. Without this address, the hardware transmitting the data would not know where to send the data.

Each data packet in the file is given a packet number. The packets of data may not have all been sent in the correct order, this will depend on the type of transmission used; you will learn more about this in Sections 2.2 and 2.3. This means that when the destination device has received all the data packets, it can use the packet number to put them back into the correct order to recreate the file.

The originators address is also normally an IP address. It is the IP address of the device from which the data has been originally sent. So, if you send data to another device, the originator's address is the address of your device. This address isn't crucial to the transmission process, but it does help to trace where data has been sent from in situations such as illegal activity, or to simply request the original device to resend the data if an error is detected.

The **payload** of the data packet is the actual data from the file that you are sending. The data is broken up into many small units to be sent as the payload in each packet.

The **trailer** section of the packet is sometimes known as the footer. This contains two main pieces of information. The first is the marker to indicate it is the end of the packet and also the data for any error detection systems that are being used. You will learn more about this in Section 2.2.

KEY WORDS

packet: a small unit of data.

packet header: a section of a packet of data that contains information about the contents of the packet and its destination.

internet protocol (IP) address: the unique address that is given to a device when it is connected to a network.

KEY WORDS

payload: the actual data that the user is sending to the receiver.

trailer: a section of a packet of data that contains information about any error checking methods that may be used.

Questions

1 Can you name the **three** main sections of a data packet?
2 What information does the packet header contain?
3 Which part of the data packet is the actual data from the file?
4 Why does data need to be broken down into packets to be transmitted?

ACTIVITY 2.1

Ask your teacher if you are allowed to use a packet tracer. Your teacher may want to complete this activity together as a class. Use the packet tracer to send data to another device and look at the data that the packet tracer software provides. What kind of information does it give you? What route did your packets take to get to the destination device?

2.2 Packet switching

You have learnt that data is broken down into packets to be sent from one device to another. The process of transmitting these packets is called **packet switching**. In this process, each packet of data is sent individually from one device to another. There could be several pathways across a **network** that the packets could be transmitted. Each data packet could be sent along a different pathway. A **router** is the device that controls which pathway will be used to transmit each packet. You will learn more about how a router does this later in the chapter. Look at the diagram of a network in Figure 2.2:

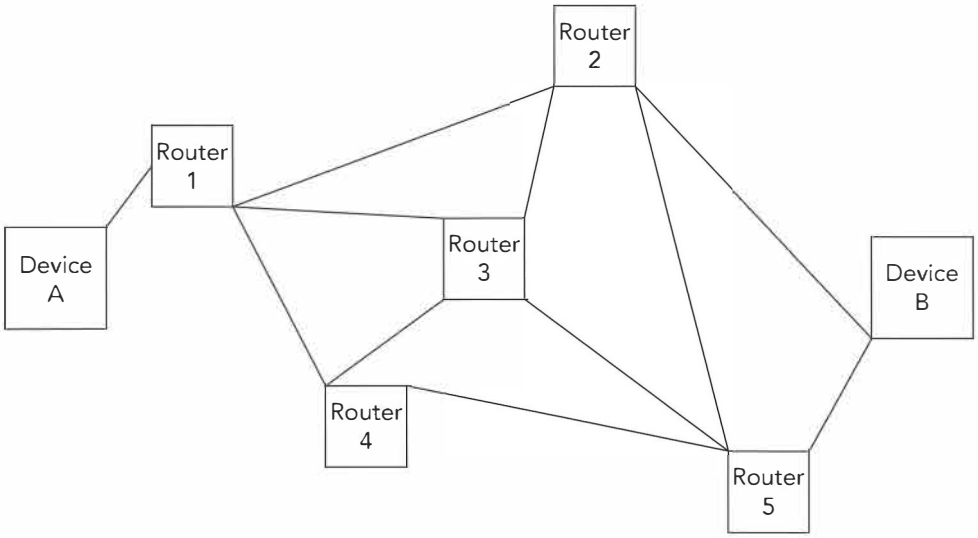


Figure 2.2: A network

The packets of data start at device A, they could be transmitted down any of the pathways between device A and device B. Each packet can be sent using a different pathway. When a packet reaches a router, the router decides which pathway to send the packet along next. This will continue until all the packets have arrived at device B. It is likely that the packets will arrive out of order. Once all packets have arrived, they are reordered to recreate the file.

KEY WORDS

packet switching: a method of transmitting data packets across a network. Each data packet is able to take an individual pathway across the network.

network: computers and devices that are joined together using cables or wireless technology.

router: a network component that examines a data packet to obtain its destination address and then forward the packet to this address.

ACTIVITY 2.2

Five packets of data need to be sent from device A to device B.

The pathway the first packet takes is:

- Router 1
- Router 2.

The pathway the second packet takes is:

- Router 1
- Router 4
- Router 5.

The pathway the third packet takes is:

- Router 1
- Router 3
- Router 2.

The pathway the fourth packet takes is:

- Router 1
- Router 4
- Router 3
- Router 5.

The pathway the fifth packet takes is:

- Router 1
- Router 3
- Router 5.

Draw a copy of the diagram of a network. Use coloured pens to draw the pathway that each of the packets of data takes.

Self Assessment

What did you do to check if your diagram is correct? Have you tried tracing each route to see if it goes through the correct routers?

2.3 Methods of data transmission

You might think that there is only one method that can be used to transmit data from one device to another, but there are actually several. You need to understand how data is transmitted using each of these methods. These include:

- serial
- parallel
- simplex
- half-duplex
- full-duplex.

Serial and **parallel** data transmission involve the number of wires that are used to transmit the data, and how many bits of data are sent at a time.

KEY WORDS

serial: a transmission method where data is sent one bit at a time down a single wire.

parallel: a transmission method where data is sent multiple bits at a time down multiple wires.

Serial data transmission

In serial data transmission, data is transmitted using a single wire. Each bit of data is transmitted one at a time along the single wire. Figure 2.3 shows an example:

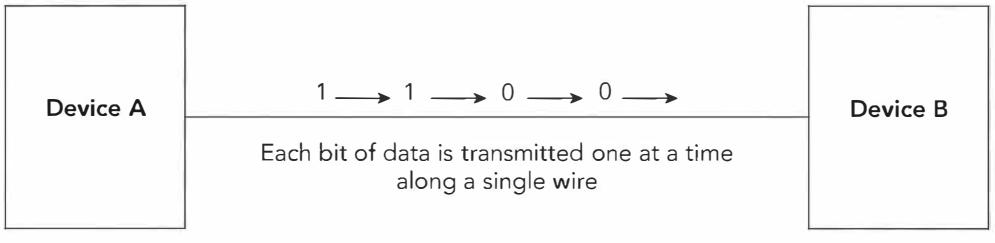


Figure 2.3: Serial data transmission

There are several advantages and disadvantages of serial data transmission, see Table 2.1:

Advantages	Disadvantages
As data is sent one bit at a time, it should arrive in order of sequence. This means there is less chance of the data being skewed.	As data is sent one bit at a time, the transmission of data is slower.
As data is sent along a single wire, there is less chance of **interference**. This means there is less chance of error in the data.	Additional data may need to be sent to indicate to the receiving device when the data transmission has started and stopped. These are called a start bit and a stop bit.
Only one wire is needed for a serial transmission cable, therefore, it is cheaper to manufacture and also cheaper to buy.	

Table 2.1: Advantages/disadvantages of serial data transmission

KEY WORD

interference: disruption, such as electromagnetism, to data when it is transmitted.

Serial transmission has less chance of data being skewed and less chance of interference. This makes it more suitable for the transmission of data over long distances.

Parallel data transmission

In parallel data transmission, data is transmitted using multiple wires. Multiple bits of data are transmitted along each wire at the same time. Figure 2.4 shows an example:

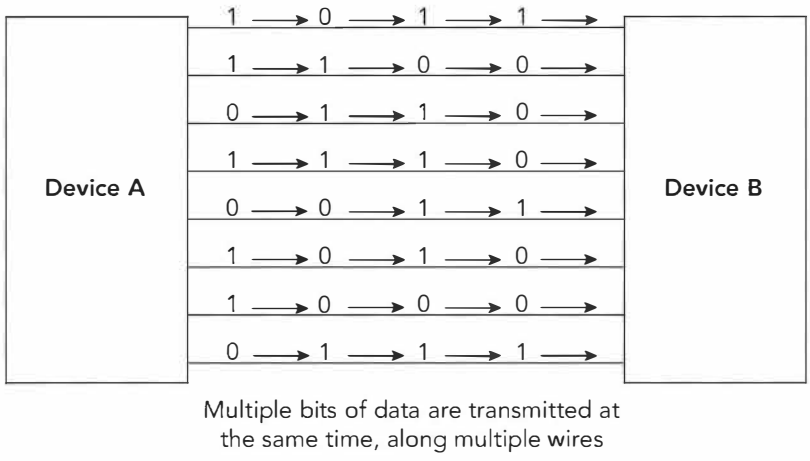


Figure 2.4: Parallel data transmission

There are several advantages and disadvantages of parallel data transmission, see Table 2.2:

Advantages	Disadvantages
As data is sent multiple bits at the same time, the transmission of data is quicker.	As data is sent multiple bits at the same time, bits do not arrive in order and need to be reordered after transmission. This increases the risk of the data being skewed.
Many computers and devices use parallel data transmission to transmit data internally. Therefore, there is no requirement to convert this to serial data transmission to transmit the data across a network.	As data is sent along multiple wires, there is more chance of interference. This means there is more chance of error in the data.
	Multiple wires are needed for a parallel transmission cable, therefore, it is more expensive to manufacture and also more expensive to buy.

Table 2.2: Advantages/disadvantages of parallel data transmission

Parallel transmission has an increased chance of data being skewed and an increased chance of interference. This makes it more suitable for the transmission of data over shorter distances. This is often limited to approximately 5 metres.

Question

5 A marketing company prints leaflets to deliver to people's houses. The leaflets are designed on a computer and sent to a large printer that is 3 metres away. Large numbers of leaflets need to be printed in a short period of time, to makes sure that they are ready for delivery. Would you use a serial or a parallel data transmission cable to connect the computer to the printer? Why would you choose this type of cable?

Simplex, **half-duplex** and **full-duplex** transmission involve the direction in which the data is transmitted.

Simplex data transmission

In simplex data transmission, data is transmitted from one device to another in one direction only. Figure 2.5 shows an example:



Figure 2.5: Simplex data transmission

You would only use this type of data transmission when there is no need to send data both ways between the devices. One example is connecting a keyboard to a computer. Simplex data transmission can be used to do this as data only needs to be sent from the keyboard to the computer, so that the computer screen can display the keys that are pressed. There is no reason why the computer would need to send data to the keyboard, so a simplex data transmission connection is used.

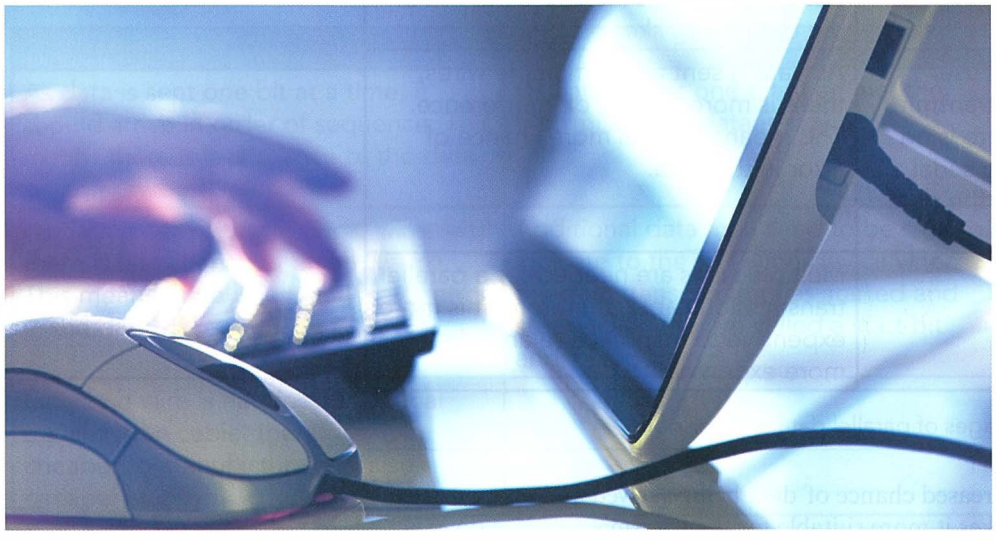


Figure 2.6: Simplex data transmission is used between a computer and a keyboard

KEY WORDS

simplex: a transmission method where data is transmitted in a single direction only.

half-duplex: a transmission method where data is transmitted in both directions, but only one direction at a time.

full-duplex: a transmission method where data is transmitted in both directions at the same time.

Half-duplex data transmission

In half-duplex data transmission, data can be transmitted in both directions between the devices, but only one direction at a time. Figure 2.7 shows an example:

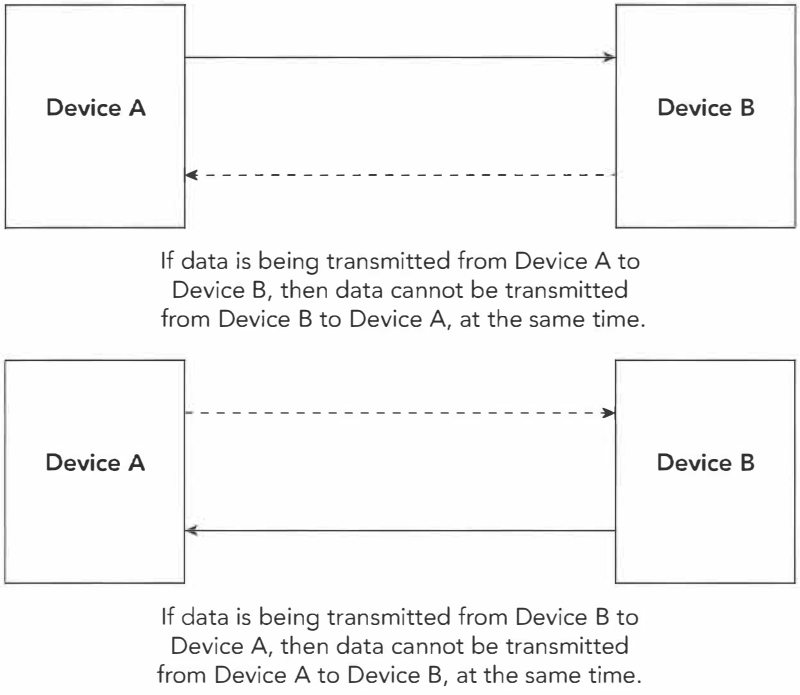


Figure 2.7: Half-duplex data transmission

You would use this type of data transmission when you need to send data in both directions between the devices, but there is no requirement for this to be at the same time. It can also be used when the communication between two devices needs to have a high level of performance. This is when two half-duplex connections can be set up between devices to allow data to be transmitted in both directions at the same time, but using two different channels of communication. Can you work out how this would be set up?

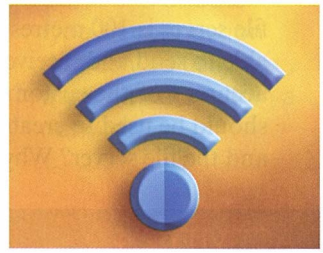


Figure 2.8: Wi-Fi uses half-duplex data transmission

ACTIVITY 2.3

Use the internet to research how half-duplex data transmission is used to create a Wi-Fi connection. Why is half-duplex, rather than full-duplex data transmission used in Wi-Fi connections?

Full-duplex data transmission

In full-duplex data transmission, data can be transmitted in both directions between the devices, at the same time. Figure 2.9 shows an example:

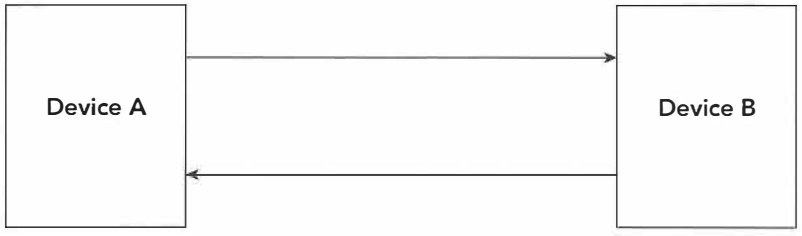


Figure 2.9: Full-duplex data transmission

You would use this type of data transmission when it is essential for data to be both sent and received by each device, at the same time. One example is a telephone conversation. The person speaking into Device A can speak at the same time as the person speaking into Device B. It may be a little bit of a chaotic conversation if you both try and speak at the same time all the time though! However, during most conversations, people do sometimes speak at the same time, so full-duplex data transmission is needed.

When a connection is created between two different devices it will either be a serial or parallel data connection. It will also be a simplex, half-duplex or full-duplex connection. This means that a connection can be, for example, a serial simplex connection, or a parallel half-duplex connection, or possibly a serial full-duplex connection. You need to understand when each of the different methods of transmission are most suitable.

Questions

6 Which method of data transmission sends data along a single wire?

7 Which method of data transmission sends data multiple bits at a time?

8 If a data transmission connection sends data one bit at a time, in both directions, but not at the same time, what kind of data transmission methods are used?

9 What are **two** advantages of serial data transmission?

10 What are **two** disadvantages of parallel data transmission?

11 A business manager transmits data about its customers to a central file server. The file server is 100 metres away from the business manager's office. They need to be able to send and receive customer data to and from the server, at the same time. The accuracy of the customer's data is imperative. Which methods of data transmission should be used to create the connection between the business manager's computer and the file server? Why would you choose those methods?

ACTIVITY 2.4

Write a data transmission scenario like the one given in Question 11. Think about which data transmission methods could be used for your scenario and why they would be the most suitable.

Peer Assessment

Give the scenario to a friend and ask them which data transmission methods they would use. Are they the same methods that you thought should be used? If you have chosen different methods, discuss and come to an agreement about which methods should be used. If you chose the same methods, see if you had the same reasons for choosing those methods.

2.4 The universal serial bus (USB) interface

The concept of **USB** can be a little confusing, as it is used to describe several aspects. You may have heard of a USB connection, a USB cable, or maybe you call a storage device 'a USB'. You need to understand the difference between several of these aspects. A USB interface includes items such as a **USB port**, a **USB cable**, a **USB connection** and a **USB device**.

A USB device, such as a USB memory stick, is a device that uses the USB interface. It plugs into a USB port on a computer and it transmits data using a USB connection. A USB cable can be used to connect devices, such as a mouse, to a computer.

A USB connection uses serial data transmission to transmit data between the devices. However, it a special type of serial data transmission connection that is used that is designed to transmit data at high speeds.

ACTIVITY 2.5

One of the issues with serial data transmission is the transmission speed is slower than parallel. Research how a USB connection is able to transmit data at a faster rate.

COMPUTER SCIENCE IN CONTEXT

Many devices are connected using a USB cable. This is because the USB interface is an industry standard, which means that manufacturers don't need to worry about compatibility issues when connecting devices. One of the most recent versions of the USB interface to be released is USB 4.

ACTIVITY 2.6

Research the data transfer speed of USB 4; is this different to USB 1, 2 and 3? What other differences can you find between the different versions of USB?

KEY WORDS

USB: an industry standard that is used to transmit data.

USB port: a socket that is a part of a device or computer that enables you to insert a USB cable.

USB cable: a type of transmission media that uses the USB method to transmit data.

USB connection: a collective name for using a USB cable plugged into a USB port to transfer data from one device to another.

USB device: the name of a device that plugs into a USB port on a computer.

ethernet: another type of connection that can be used to transmit data within a network.

There are several advantages and disadvantages of the USB interface, see Table 2.3:

Advantages	Disadvantages
It is a simple interface. The USB cable to device can only fit into the USB one way. Therefore, it means less errors in connecting devices are likely to be made.	The length of a USB cable is limited, normally to 5 metres.
The speed of a USB connection is relatively high, so data can be transferred quickly.	The transmission speed is relatively high for a USB connection, but it isn't as high as other types of connection, such as **ethernet**.

(continued)

Advantages	Disadvantages
The USB interface is a universal standard for connecting devices, therefore a USB port is included in many different devices.	
When a USB device to cable is inserted into a USB port, it is automatically detected. The first connection will normally involve the download of drivers to operate the hardware that has been connected. Each connection after this the driver should not need to be downloaded again.	
A USB connection can also be used to power a device, so it does not need another power source. It can also use this to charge a device, such as a mobile phone.	

Table 2.3: Advantages/disadvantages of the USB interface

Questions

12 What does USB stand for?

13 What type of data transmission method does a USB interface use?

14 What are **two** advantages of using the USB interface?

15 Can you name **five** devices that are connected using a USB interface?

16 What is the length a USB cable limited to? Can you think why it is limited to this length?

2.5 Detecting errors in data transmission

The transmission of data from one device to another is often not a perfect process. In the process of transmitting the data, interference can occur. This can cause data to be lost, data to be gained and data to change. This would not be helpful and could also lead to many issues. If a password wasn't transmitted correctly, then that person may not be able to log into their account. If a person's address wasn't transmitted correctly, they may not get email that needs to be sent to them. If a person's details are not transmitted correctly to a company, they may not get a product or service that they order. The accuracy of data is often vital, therefore there needs to be a procedure in place to detect any errors in data so that actions can be taken to correct this.

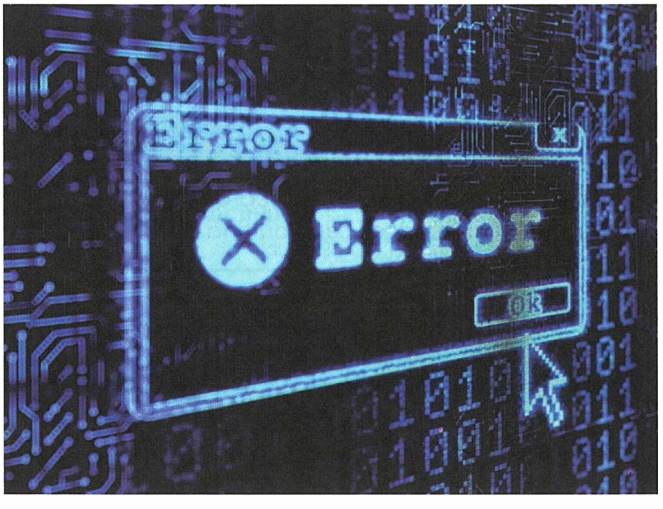


Figure 2.10: Procedures need to be in place to handle errors in data transmission

There are several error detection methods that can be used, these include:

- parity check
- checksum
- echo check.

Each of these methods is designed to check for errors after the data has been transmitted from one device to another.

Parity check

A **parity check** can use an odd or even check method. Each byte of data has 7 bits and 1 extra bit that is called a parity bit (see Figure 2.11). The parity bit is normally the first or last bit of data in the byte.

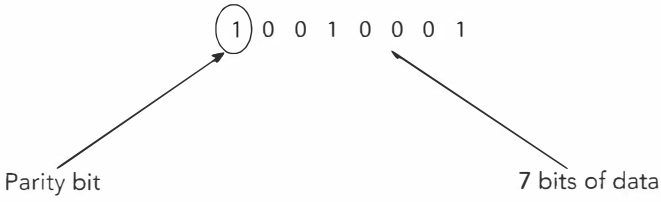


Figure 2.11: The parity bit is the first bit of data in this byte

KEY WORD

parity check: a type of error detection method that adds an additional bit to each byte to create an odd or even sum.

Before transmission begins, the parity check is set to be either odd or even parity. The number of 1s in the 7 bits of data is totalled. In the example in Figure 2.11, the result of that would be 2. If an odd parity check is used, then a 1 is added as a parity bit. This is because all the 1s in the byte now add up to 3, which is an odd number. If an even parity check is used a parity bit of 0 would have been added instead. This is because all the 1s in the byte would then add up to 2, which is an even number. When the parity bit has been added to each byte, the data can be transmitted. After transmission, the receiving device will check each byte of data for errors. If an odd parity check has been used and the device finds a byte that has an even number of 1s, then it knows that an error has occurred with this byte of data. The error is detected!

Questions

17 Which of these bytes would have been transmitted incorrectly if an even parity check has been used?

- **a** 10111011
- **b** 01110111
- **c** 10101000

18 Which of these bytes would have been transmitted incorrectly if an odd parity check has been used?

- **a** 00110001
- **b** 10110101
- **c** 10001000

19 Can you think of an instance in which an error could occur in the data, but this would not be detected by a parity check?

Checksum

A **checksum** uses a calculated value to check for errors. A value is calculated from the data that will be transmitted, before transmission takes place. A method such as modulus 11 could be used to calculate the value.

KEY WORD

checksum: a type of error detection method that performs a calculation on the data to create a checksum value. Checksum values are compared after transmission to see if they match.

ACTIVITY 2.7

Use the internet to find out how the modulus 11 method calculates a value from the data. You do not need to be able to remember how to do this method of calculation for the exam, but it will help you to know how a checksum is calculated.

Once the checksum value has been calculated it is added to the data to be transmitted with it. After transmission, the receiving device uses the same method to calculate a value from the received data. If the values match, then the device knows that no error has occurred during transmission. If the values do not match, the device knows that an error has occurred during transmission. The error is detected!

Questions

20 What is the checksum value calculated from?

21 What happens to the checksum value when it has been calculated before transmission?

22 Why would checksum values that do not match show that an error has occurred?

Echo check

An **echo check** involves a simple comparison of the data sent to the data received. The sending device transmits the data to the receiving device. The receiving device then transmits the data it receives back to the sending device. The sending device compares the data it sent to the data it has received back from the receiving device to see if they match. If they do, then no error has occurred. If they don't match, then the sending device knows the data was received with error. The error is detected!

KEY WORD

echo check: a type of error detection method that sends a copy of the transmitted data back to the sender to be compared with the original data sent.

ACTIVITY 2.8

Draw a diagram to represent how an echo check detects errors in data.

Question

23 If the data does not match that which is sent back by the receiving device, will this always mean that the receiving device did not receive the data correctly? How do you know that?

Automatic repeat request (ARQ)

When an error has been detected after the data is transmitted, it is likely that the data will need to be retransmitted. Either the sending device will need to ask the receiving device if it received the data correctly, or the receiving device will need to tell the sending device it did or did not receive the data correctly. This will allow the data to be retransmitted if necessary.

A method called **automatic repeat request (ARQ)** can be used to do this. There are two main ways that an ARQ can operate and each method uses either a positive or negative **acknowledgement** and a **timeout**.

In a positive acknowledgement method:

- The sending device transmits the first data packet.
- The receiving device receives the data and checks it for errors.
- Once the receiving device knows it has received the data error free, it sends a positive acknowledgement back to the sending device.
- When the sending device receives this positive acknowledgement, it knows the receiving device has received the data packet error free and it sends the next data packet.
- If the sending device does not receive a positive acknowledgement within a set timeframe, a timeout occurs.
- When a timeout occurs, the sending device will resend the data packet. It will keep doing this when a timeout occurs, until it receives a positive acknowledgement, or sometimes a limit (such as 20 times) is set and when this limit is reached it will stop resending the data.

ACTIVITY 2.9

Draw a diagram to represent how an ARQ system operates using positive acknowledgement.

In a negative acknowledgement method:

- The sending device transmits the first data packet.
- The receiving device receives the data packet and checks it for errors.
- If the receiving device detects no errors, no further action is taken.
- If the receiving device does detect errors, it will send a negative acknowledgement back to the sender.
- If the sender receives a negative acknowledgement, it knows this means the data was received incorrectly, so it can resend the data packet.
- A timeout is set by the sending device when it sends the data. This is just so that the sending device knows that if it doesn't receive a negative acknowledgement back within that set time period, it doesn't need to be still be waiting for it and can send the next data packet.

KEY WORDS

automatic repeat request (ARQ): a type of error detection method that uses acknowledgement and timeout to see if data has arrived correctly after transmission.

acknowledgement: a message that is sent from one device to another to indicate whether data is received correctly.

timeout: a period of time that is set and used to wait for an acknowledgement to be received.

ACTIVITY 2.10

Draw a diagram to represent how an ARQ system operates using negative acknowledgement.

REFLECTION

After completing activities 2.8, 2.9 and 2.10:
How have you made sure that your diagrams for an echo check and both ARQ methods are correct?

Check digit

You have learnt that errors, such as data loss, data gain and data being changed can occur when data is transmitted. You now know that these errors normally occur due to interference. It is also possible for errors to occur when data entry is performed. This could be manual data entry, for example, a human typing in a value, or it could be automatic data entry, for example, a barcode scanner scanning a barcode to obtain the data stored in the barcode.

A method is necessary to check for errors with data entry and this method is called a **check digit**. This is how a check digit operates:

- A check digit value is previously calculated from the data that will be entered at some point, for example, a barcode number or an ISBN number. This number is stored with the data.
- When the data is entered, for example, the barcode is scanned, the check digit is recalculated from the data entered.
- If the previously calculated check digit and the stored check digit match, the data entered is correct.
- If the previously calculated check digit and the stored check digit do not match, the data entered is incorrect.

Although the process is similar, you should not confuse the operation of a check digit and a checksum. Make sure that you take the time to understand the difference between the two.

KEY WORD

check digit: a type of error detection method that is used for data entry. A calculation is performed on the data entered to create a value. Check digit values are compared to see if the data entered is correct.

ACTIVITY 2.11

With a partner, discuss and write down the differences between a check digit and a checksum.

COMPUTER SCIENCE IN CONTEXT

Companies will normally combine several error checking methods to make sure that the data they transmit and receive is error free. This means that the data can be checked in several ways to make sure that any error that could occur is being checked. This is because the data that many companies use is vital to their business, they want to do everything they can to make sure that it does not contain any errors caused by transmitting the data.

2.6 Encryption

You have learnt a lot about how data is transmitted in this chapter. You have also learnt why it is important for data to be received correctly and how errors can be detected to make sure this happens. You also need to understand that the data that is transmitted often needs to be protected during transmission. This is because data is valuable, especially when it contains our personal details and information that is personal to us. A hacker may try and steal the data during transmission so that they can use it for their own gain. One method of protection that can be used when transmitting data is **encryption**.

Data before encryption takes place it is called **plain text**. An encryption algorithm, called an **encryption key**, is used to scramble the data and make it meaningless. This meaningless data is called **cipher text**. The cipher text can then be transmitted from one device to another. The receiving device uses the key to decrypt the cipher text and return it to its plain text form. This means the data will have meaning again if read.

One important thing to note about encryption is that is does not stop a hacker stealing the data that is transmitted. It just means that the data that is stolen will be meaningless, as it will be encrypted and therefore scrambled. They will not understand any of the data in the file.

There are two main methods of encryption that can be used **symmetric** and **asymmetric**.

The process for symmetric encryption is:

- Plain text is encrypted into cipher text using an encryption key.
- The cipher text and the encryption key are sent separately to the receiving device.
- The same key is then used to decrypt the cipher text back into its plain text form.

This is the simplest method of encryption and is still used for data that is not of a high level of importance. It is sometimes used by people just wanting to encrypt the data that they store on their own computer or external storage device, as it isn't being transmitted across a network. However, people began to worry that it could be too easy for a hacker to intercept both the cipher data and the encryption key if it is sent across a network, or the internet, so a more secure method was developed called asymmetric encryption.

The process for asymmetric encryption is:

- Plain text is encrypted into cipher text using a public key. This is also a type of encryption algorithm.
- The cipher text is transmitted to the receiving device.
- The cipher text cannot be decrypted using the public key, it is decrypted using a private key.

A person who uses an asymmetric method of encryption has a public encryption key. This is called a public key because it is made public for the encryption process. Making something public means that anyone can see and use it. Any device that wants to send you encrypted data may obtain your public key and encrypt the data using your public key. You also have a private key. Any encrypted data that your device receives that has been encrypted with your public key can be decrypted with your private key. They act as a pair. Your private key is always kept private though, your device does not make it available to any other unless authorisation is given to do so.

KEY WORDS

encryption: a method of securing data for storage or transmission that scrambles it and makes it meaningless.

plain text: the name given to data before encryption.

encryption key: a type of algorithm that is used to encrypt data.

cipher text: the name given to data after transmission.

symmetric: a type of encryption that uses the same key to encrypt and decrypt data.

asymmetric: a type of encryption that uses two different keys to encrypt and decrypt data.

The process can work in reverse as well (a private key is used to decrypt and a public key is used to decrypt), and this is what can occur in processes such as digital signatures. These are used in security protocols such as SSL (secure sockets layer). You will learn more about these in Chapter 5, about applications of the internet.

COMPUTER SCIENCE IN CONTEXT

A bank uses asymmetric encryption for all the banking transactions that are completed on a daily basis. As you can imagine, the data that is transmitted for these banking transactions is very sensitive and personal data, so the bank has to make sure that they use a very secure method of encryption for this process.

Questions

24 What is the name given to an encryption algorithm?

25 What is the name given to data that has been encrypted?

26 Does encryption stop data being stolen by a hacker? Why or why not?

27 Why is asymmetric encryption a more secure method of encryption than symmetric?

SUMMARY

Data is broken down into packets to be transmitted from one device to another.
Each packet contains three parts: the packet header, the payload and the trailer.
The packet header includes the destination address, the packet number and the originator's address.
The payload is the data the sender wants to transmit.
The trailer contains data such as the error detection method to be used.
A process called packet switching can be used to send the data from one device to another across a network.
Data is transmitted using serial or parallel transmission.
Data is also transmitted using simplex, half-duplex or full-duplex transmission.
An interface called USB can be used to transmit data. This is often used to connect hardware such as a keyboard to a computer.
Errors can occur when transmitting data due to interference.
Methods are required to detect any errors in transmission.
A parity check is an error detection method that uses a parity bit to detect errors. An odd or even parity check method can be used.
A checksum is an error detection method that uses a calculated value to detect errors.
An echo check compares data that is sent and received to see if they match, to detect errors.

CONTINUED

An ARQ can be used to monitor whether data is received correctly after transmission.
A check digit is an error detection method that is used for data entry.
Data can be encrypted to keep it secure during transmission.
Encryption can be performed using a symmetric or asymmetric method.
Symmetric encryption uses the same key to encrypt and decrypt the data.
Asymmetric encryption uses different key, a public and a private key, to encrypt and decrypt the data.

EXAM-STYLE QUESTIONS

1 **Identify** *two* items of data that would be included in the header of a packet of data. **[2]**

2 **Explain** why and how packets of data are reordered after packet switching has been used to transmit data across a network. **[2]**

3 **Describe** how data is transmitted using serial half-duplex data transmission. **[4]**

4 **Give** *two* benefits of using the USB interface to connect hardware devices to a computer. **[2]**

5 A company has a central file server that is located 500 m away from the main office where employees work at their computers.

Employees send and retrieve files to and from the file server on a daily basis, often at the same time.

Identify the type of data transmission that would be most suitable for the given context. Explain why this would be the most suitable method of transmission. **[6]**

6 Four statements are given about error detection methods.

Tick (✓) to show which statement applies to which error detection method. Some statements may apply to more than one error detection method. **[4]**

Statement	Checksum	Echo check	Parity check
This method checks for errors in data after it has been transmitted.			
This method can use an odd or even process.			
This method sends data back from the receiver to the sender to compare the data to check for errors.			
This method uses a value that is calculated from the data to check for errors.			

COMMAND WORDS

identify: name / select / recognise.

explain: set out purposes or reasons / make the relationships between things evident / provide why and / or how and support with relevant evidence.

describe: state the points of a topic / give characteristics and main features.

give: produce an answer from a given source or recall / memory.

CONTINUED

7 The given paragraph describes how data is encrypted using symmetric encryption.

Complete the missing terms in the paragraph using the given list.
Not all terms need to be used.

- a different
- cipher text
- decrypted
- encrypted
- key
- lock
- meaningless
- plain text
- the same
- understandable

Before encryption the data is known as ..... . This data is encrypted using an algorithm that is known as a ..... .
After the data has been encrypted it is known as ..... .
This type of data is scrambled and ..... if it is stolen.
When the data is received it is ..... using ..... encryption algorithm. **[6]**

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics.
This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising.
You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
describe the structure of a packet of data.	2.1			
describe the process of packet switching.	2.2			
describe how data is transmitted using serial and parallel data transmission.	2.3			
describe how data is transmitted using simplex, half-duplex and full-duplex data transmission.	2.3			
choose a suitable data transmission for a given context and explain why that is the best method of data transmission.	2.3			

CONTINUED

I can...	See topic	Needs more work	Getting there	Confident to move on
describe how data is transmitted using the USB interface.	2.4			
explain how errors can occur during data transmission.	2.5			
describe how a parity check uses a parity bit to detects errors.	2.5			
describe how a checksum value can be used to detect errors.	2.5			
describe how an echo check compares data to detects errors.	2.5			
describe how an ARQ can monitor if data is received correctly.	2.5			
describe how a check digit can be used to detect errors involving data entry.	2.5			
explain why data may need to be encrypted.	2.6			
describe how data is encrypted using symmetric encryption.	2.6			
describe how data is encrypted using asymmetric encryption.	2.6			